

1. Find the mean and the median of: 88, 72, 33, 29, 70, 54, 86, 91, 57, 61.

Ans [mean=64.1, median= 65.5].

2. The numbers 3.2, 7.9, 4.5 have frequencies f , $(f+2)$, $(f-2)$ and $(f+6)$ respectively. If the arithmetic mean is 4.876, find the value of f .

Ans[5].

3. A factory has five sections employing 105, 184, 130, 93, 124 workers. The mean earnings in certain week per worker are Rs. 363.84, 365.12, 365.27, 368.19, 364.22 for five sections. Find out the mean earnings per worker for the whole factory.

Ans [Rs. 365.21].

4. Calculate the median from the following data:

Value	Frequency
<10	4
<20	16
<30	40
<40	76
<50	96
<60	112
<70	120
<80	125

Ans [36.25].

5. In a group of 1000 people, the monthly income of 4% people are below Rs. 60 K, 15% under Rs. 62.5 K, 15% earn 95 K or above and 5% get Rs 100 K or above. Find the

median income.

Ans [Rs. 78.75 K.]

6. Suppose the data set is $\{10, 3, 1, 7, 12, 14, 5, 15, 6, 16, 13, 17\}$. Find out the quantile of order $\frac{1}{3}$.

Ans [6.3].

7. The standard deviation of a set of 32 observations is 5. If the sum of the observations is 80, what is the sum of the squares of these observations?

Ans [1000].

8. For a data set of 280 observations, mean is 54 and standard deviation is 3. If two observations with value 64 and 80 are replaced by 62 and 82, respectively, calculate the new mean and standard deviation.

Ans [mean 54, sd 3.04].

9. From some financial statistics, it is found that the monthly average electricity charges was Rs. 2460 and sd Rs. 120. The monthly average direct wages was Rs. 42000 and sd Rs. 1200. State which is the more variable with proper reason.

Ans [since the cv of electricity charges is larger than cv of direct wages, electricity charges are more variable].

10. The first four moments of a data set about the value 3 are 2, 10, 40, 218. Find the first four moments about origin and mean.

Ans [first four moments about origin: 5, 31, 211, 1535; first four moments about mean: 0, 6, -4, 90].

11. The first four moments of a distribution about the value 4 of the variables are -1.5, 17, -30 and 108. Find the values of beta coefficients.

Ans [$\beta_1 = 0.492, \beta_2 = 0.654$].

12. The arithmetic mean of a distribution is 5. The second and third moments about the mean are 20 and 140 respectively. Find the third moment of the distribution about 10.

Ans [-285].

13. The first three moments of a distribution about the value 3 of the variable are 2, 10, 30 respectively. The first three moments about zero are respectively.

Ans[5, 31, 201]

14. Sum of 10 numbers is 80 and sum of their square is 650. Find the coefficient of variation.

Ans[12.5] %

15. If $\bar{x}_1 = \frac{1}{n} \sum_{i=1}^n x_i$, $\bar{x}_2 = \frac{1}{n} \sum_{i=2}^{n+1} x_i$ and $\bar{x}_3 = \frac{1}{n} \sum_{i=3}^{n+2} x_i$, then show that

(i). $\bar{x}_2 = \bar{x}_1 + \frac{1}{n}(x_{n+1} - x_1)$, and (ii). $\bar{x}_3 = \bar{x}_2 + \frac{1}{n}(x_{n+2} - x_2)$.

16. In a series of measurements we obtain m_1 values of magnitude x_1 , m_2 values of magnitude x_2 and so on. If $\bar{x}$ is the mean value of all the measurements, prove that the standard deviation is

$$\sqrt{\frac{\sum m_r (k - x_r)^2}{\sum m_r} - \delta^2}, \text{ where } \bar{x} = k + \delta \text{ and } k \text{ is a constant.}$$

17. Let r be the range and $S = \left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{1/2}$, be the standard deviation of a set of observations $x_1, x_2, \dots, x_n$ then prove that
- i). $S \leq r \left(\frac{n}{n-1} \right)^{1/2}$ and ii). $S \leq r$
18. Show that, if the variable takes the value $0, 1, 2, \dots, n$ with frequencies proportional to the binomial coefficients ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$, respectively then the mean of the distribution is $(n/2)$, the mean squared deviation about $x = 0$ is $\frac{n(n+1)}{4}$ and the variance is $\frac{n}{4}$.